Lecture 01: Part b

Reducible to separable form:

1. Solve: $\frac{dy}{dx} = (x+y)^2$ —— (1)

Let

$$x+y = u$$

$$\therefore 1 + \frac{dy}{dx} = \frac{du}{dx} \Rightarrow \frac{dy}{dx} = \frac{du}{dx} - 1$$

$$(1) \Rightarrow \frac{du}{dx} - 1 = u^2$$

$$\Rightarrow \frac{du}{dx} = (1+u^2) \Rightarrow \frac{du}{1+u^2} = dx$$

Integrating, we get

$$\int \frac{du}{1+u^2} = \int dx \Rightarrow \tan^{-1} u = x + c$$

$$\Rightarrow \boxed{\tan^{-1}(x+y) = x + c}$$

Ans

2. Solve: $\sin^{-1}\left(\frac{dy}{dx}\right) = x+y$

Ans: $\sin^{-1}\left(\frac{dy}{dx}\right) = (x+y)$

$$\Rightarrow \frac{dy}{dx} = \sin(x+y) \text{ —— (1)}$$

Let

$$x+y=u \Rightarrow \therefore \; 1+\frac{dy}{dx}=\frac{du}{dx}$$

$$\Rightarrow \frac{dy}{dx}=\frac{du}{dx}-1$$

1 becomes

$$\frac{du}{dx}-1=\sin u$$

$$\Rightarrow \frac{du}{dx}=1+\sin u \Rightarrow \frac{du}{1+\sin u}=dx$$

Integrating, $\int \frac{du}{1+\sin u}=\int dx$

$$\Rightarrow \int \frac{1-\sin u}{(1+\sin u)(1-\sin u)}\,du=x+c$$

$$\Rightarrow \int \frac{1-\sin u}{1-\sin^2 u}\,du=x+c$$

$$\Rightarrow \int \frac{1-\sin u}{\cos^2 u}\,du=x+c$$

$$\Rightarrow \int \sec^2 u\,du-\int \sec u\tan u\,du=x+c$$

$$\Rightarrow \tan u-\sec u=x+c$$

$$\Rightarrow \tan(x+y)-\sec(x+y)=x+c$$

Example 3: Solve $\cos(x+y)\,dy = dx$

Ans:

$$\frac{dy}{dx} = \sec(x+y)$$

Let

$x+y = u$

$$\frac{dy}{dx} = \frac{du}{dx} - 1$$

$$\Rightarrow \frac{du}{dx} - 1 = \sec u$$

$$\Rightarrow \frac{du}{dx} = 1 + \sec u$$

$$\Rightarrow \int \frac{du}{1+\sec u} = \int dx$$

$$\Rightarrow \int \frac{du}{1+\frac{1}{\cos u}} = x + c$$

$$\Rightarrow \int \frac{du}{\frac{\cos u + 1}{\cos u}} = x + c$$

$$\Rightarrow \int \frac{\cos u}{1+\cos u}\,du = x + c$$

Division

$$\begin{array}{r|l|l} 1+\cos u & \cos u & 1 \\ & \cos u + 1 & \\ \hline & (-) \quad -1 & \end{array}$$

$$\therefore \frac{\cos u}{1+\cos u} = 1 - \frac{1}{1+\cos u}$$

$$\Rightarrow \int \left(1 - \frac{1}{1+\cos u}\right) du = x + c$$

$$\Rightarrow \int \left(1 - \frac{1}{2\cos^2\frac{u}{2}}\right) du = x + c$$

$$\Rightarrow \int \left(1 - \frac{1}{2}\sec^2\frac{u}{2}\right) du = x + c$$

$$\Rightarrow u - \frac{\cancel{1}}{\cancel{2}} \frac{\tan\frac{u}{2}}{\cancel{\frac{1}{2}}} = x + c$$

$$\Rightarrow (x+y) - \tan\left(\frac{x+y}{2}\right) = x + c$$

4 Solve $\frac{dy}{dx} = \frac{1-x-y}{x+y}$

$\Rightarrow \frac{dy}{dx} = \frac{1-(x+y)}{x+y}$

Let $x+y = u \therefore \frac{dy}{dx} = \frac{du}{dx} - 1$

$\Rightarrow \frac{du}{dx} - 1 = \frac{1-u}{u}$ (*) $\Rightarrow \frac{du}{dx} = 1 + \frac{1-u}{u}$

$= \frac{\cancel{u}+1-\cancel{u}}{u}$

$\Rightarrow \int u \, du = \int dx$

$\Rightarrow \frac{u^2}{2} = x + c$

$\Rightarrow \frac{(x+y)^2}{2} = x + c$

Home task:

Solve: 1. $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$

2. $\frac{dy}{dx} = \cos(x+y)$

3. ** $\frac{dy}{dx} = \tan(x+y)$ [MAT 130]